Week 3: SQL & Data Querying — Queries and Insights

AnalystLab Africa Data Analytics Internship

Part 1: Chinook Database (Music Store Data)

Database Setup

The Chinook database was accessed using MySQL, with SHOW DATABASES, USE CHINOOK, and SHOW TABLES used to verify the database and available tables. The analysis focused mainly on the customer and invoice tables, which were used to examine customer information and transaction data.

Core SQL Queries

SQL filtering, sorting, grouping, and aggregation were applied to analyse the customer base and revenue. Customers from Brazil were filtered and sorted by state using WHERE and ORDER BY. Customers were also grouped by country using GROUP BY and counted using COUNT(). The HAVING clause was used to identify countries with more than four customers. The results showed that the **USA had the highest number of customers (13)**, followed by **Canada (8)**, while **France and Brazil had 5 each**, across **24 countries**. SUM() and AVG() were also used to calculate total revenue and average invoice value.

Advanced SQL Concepts

An INNER JOIN between the customer and invoice tables was used to calculate total customer spending. **Helena Holý was identified as the highest-spending customer, with total purchases of $49.62**. A LEFT JOIN was used to identify customers without invoices, but the query returned **zero records**, indicating that every customer in the dataset had made at least one purchase.

Subqueries were used to calculate total spending per customer and determine the average customer spending. The average was approximately **$39.47**, and a further query identified **22 customers** whose spending exceeded this average. These customers represent a valuable group that could be targeted through customer retention and loyalty strategies.

Window functions were applied using ROW_NUMBER() and RANK() to analyse invoice values. PARTITION BY customerid was used with ROW_NUMBER() to rank each customer's invoices independently and identify their highest-value invoice. RANK() was also used to rank invoices by total value while allowing equal values to share the same rank.

Query Optimization

An index was created on invoice.customerid because the column is frequently used in joins and customer-level analysis. Although the Chinook dataset is relatively small, indexing this column can improve query performance when working with larger transactional datasets.

Key Insights

- The **USA leads the customer base with 13 customers**, followed by Canada with 8.
- Customers are distributed across **24 countries**, indicating a geographically diverse customer base.

- **Helena Holý** was the highest-spending customer, with total purchases of **$49.62**.
- Every customer in the dataset had made at least one purchase, with **no customers identified without invoices**.
- The average customer spending was approximately **$39.47**, and **22 customers spent above this average**.
- ROW_NUMBER() and RANK() provided useful methods for analysing transaction rankings and customer purchasing behaviour.
- Indexing customerid demonstrates a basic optimisation technique that can support better query performance as data volume increases.